

AYMANE AIT DADS

Generative AI & Agentic AI Intern | LLM Fine-Tuning · Agentic Pipelines · Data Science

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PROFESSIONAL SUMMARY

Led data science work at Orange Maroc processing 100K+ fiber-optic network records with Random Forest and K-Means models, reducing manual analysis time by ~60% and delivering actionable maintenance recommendations directly to a network optimization team. Ranked 17th out of 3,000+ teams in the ongoing NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by fine-tuning a 30B-parameter model with LoRA SFT, demonstrating depth in Generative AI and large language model development. Brings hands-on experience in agentic pipelines, LLM orchestration, RAG, and prompt engineering - skills that map directly to Capgemini's SAP Technology cluster's goal of deploying AI agents to support developers and consultants. Aims to translate these capabilities into concrete, responsible Generative AI and Agentic AI use cases within SAP-integrated workflows at Capgemini Utrecht.

CORE COMPETENCIES

Generative AI | Agentic AI | LLM Orchestration | Prompt Engineering | Data Science | Software Integration | Responsible AI | Actionable Insights

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built Random Forest and K-Means classification models on **100K+ fiber-optic network records** (upload speed, latency, jitter) to assess quality across thousands of network nodes.
- Developed a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for operational decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation pipelines built with Python, Pandas, and Scikit-learn.
- Delivered stakeholder presentations in **Power BI** translating complex model outputs into concrete maintenance recommendations, bridging technical findings with business decision-making.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge (In Progress)

Kaggle · 2025-2026 · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in an ongoing Generative AI reasoning competition by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers to generate verified chain-of-thought training data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM

AI-Generated Image Detection

EURECOM + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in the EURECOM class private Kaggle leaderboard** (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M params) with a custom MLP head, dual learning rates, and 10-view TTA across 250K images from 25+ generator types.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; identified that 1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC), demonstrating stronger out-of-distribution generalization with reduced overfitting.

PyTorch · OpenCLIP · ViT · AdamW · FP16

Twitter Sentiment Analysis

EURECOM NLP Course · 2025

- Built an end-to-end NLP pipeline covering tokenization, **TF-IDF, Word2Vec, and transformer fine-tuning** for sentiment classification, applying hyperparameter optimization to maximize classification accuracy.
- Applied text preprocessing, embedding strategies, and **agentic prompt engineering** principles to compare classical and transformer-based approaches, producing actionable insights on model selection for business NLP tasks.

Python · Hugging Face Transformers · Scikit-learn · Pandas

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Web Applications, Statistics, Image Security

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Data Science & Analytics: Python | SQL | Pandas | NumPy | Power BI | Scikit-learn | Feature Engineering

Generative AI & LLM Engineering: LoRA / QLoRA | Hugging Face Transformers | LangChain | RAG | Prompt Engineering | Agentic Pipelines | vLLM

MLOps & Infrastructure: PyTorch | Git | Docker | Supabase | REST APIs | FP16/BF16 Mixed Precision | Linux